

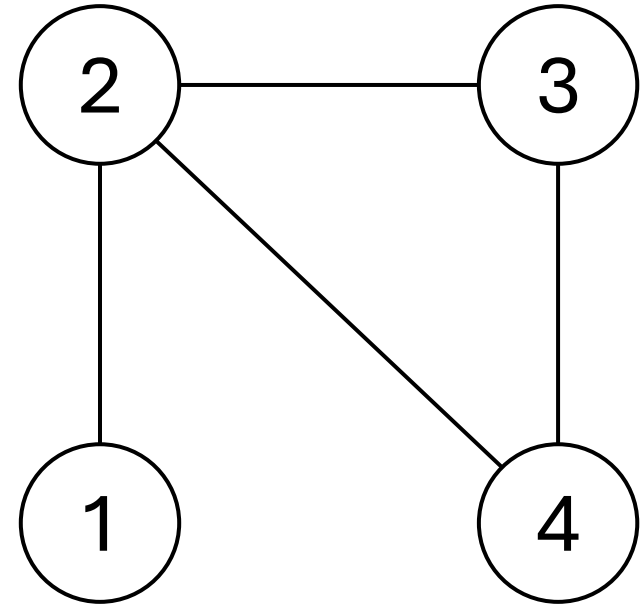


IFORS 2026
The 24th Conference of the International Federation
of Operational Research Societies
12-17 JULY
VIENNA, AUSTRIA

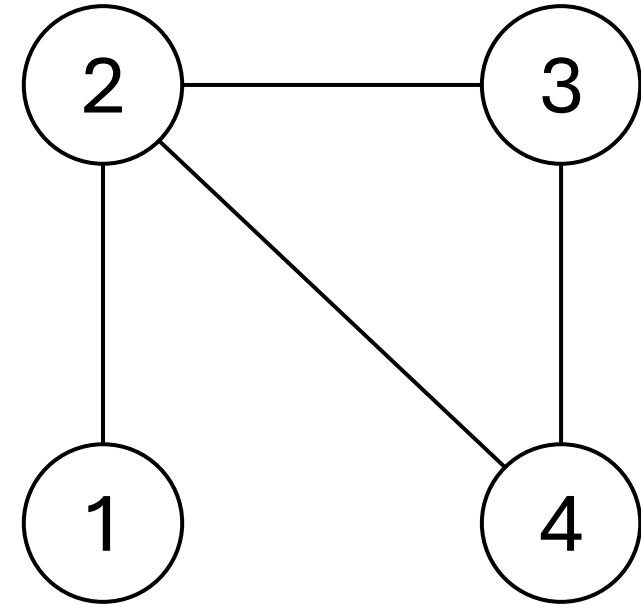
Dynamic matching models under greedy policies: performance paradox and extensions to hypergraphs

Ana Bušić, Josu Doncel, Jean-Michel Fourneau
Shu Li, Doan-Dai Nguyen
INRIA Paris & École Normale Supérieure - PSL

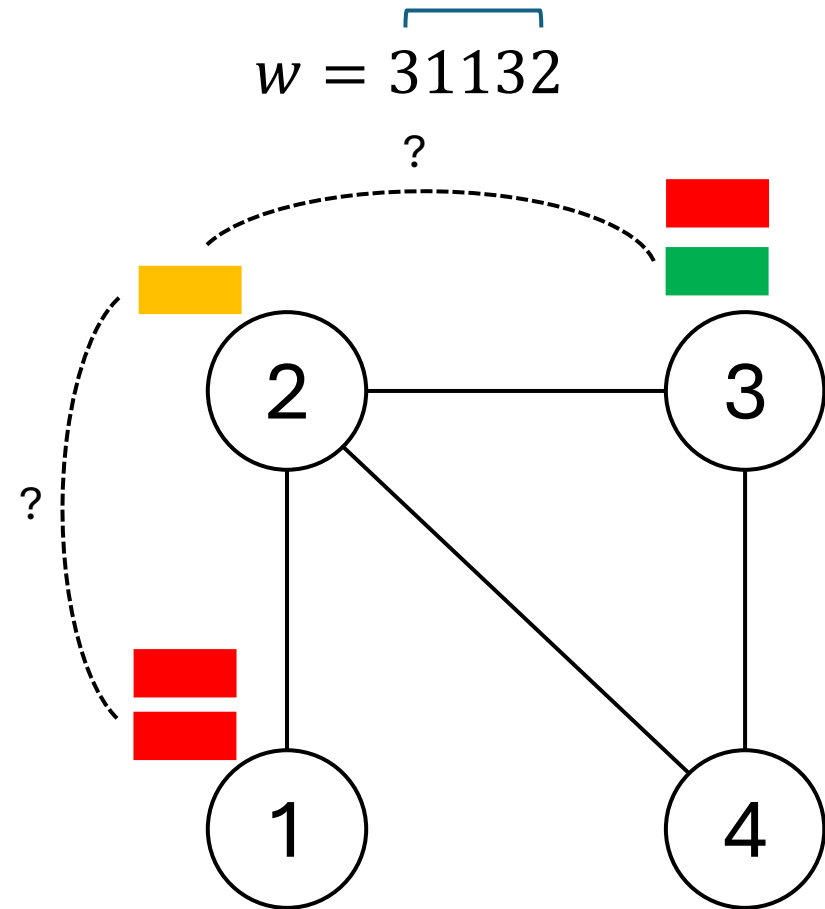
- Motivation: kidney paired donation.
- Other application: book swap, home swap, etc.
- Features:
 1. Compatibility
 2. Random arrivals
 3. Finite waiting time



1. Compatibility: An undirected graph $G = (V, E)$
 - Vertices u, v : item classes;
 - Edges $e = (u, v)$: matching types.
Two items of class u and v in the buffer *can* form a type- e matching, then leave the buffer.
2. Random arrivals:
 - Distribution μ on V .
 - At time t , a class- v item arrives to the buffer i.i.d. according to μ .
3. Finite waiting time: Stability
 - Under Markovian policies, the buffer is a Markov chain $X(t)$.
 - Stability = X positive recurrent.



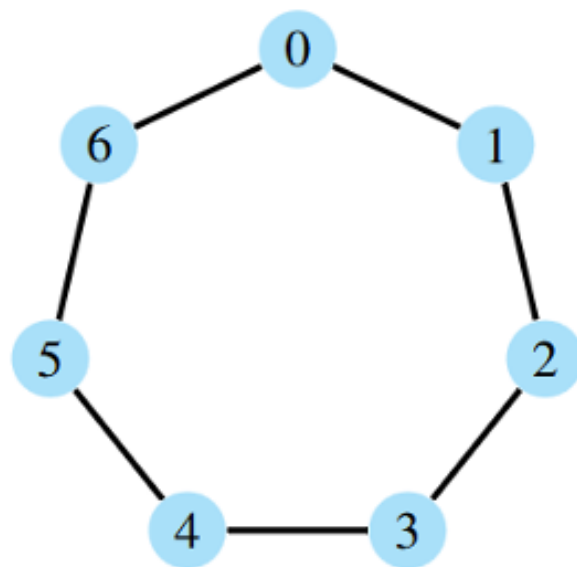
- There can be multiple matchings, and a matching policy is needed.
 - Greedy policies = matching if possible
- First-Come-First-Matched (FCFM):
 - Greedy
 - Always stabilising
 - Perceived as fair
 - Admit a product-form stationary distribution, but very hard to compute.



$$\pi(w) = \frac{1}{C} \prod_{i=1}^d \frac{\mu(w_i)}{\mu(N(\{w_1, \dots, w_i\}))}$$

$$C = 1 + \sum_{\text{U-independent}} \sum_{\sigma \in \mathcal{S}(|U|)} \prod_{j=1}^{|U|} \frac{\mu(u_{\sigma(j)})}{\mu(N(\{u_{\sigma(1)}, \dots, u_{\sigma(j)}\})) - \mu(\{u_{\sigma(1)}, \dots, u_{\sigma(j)}\})}$$

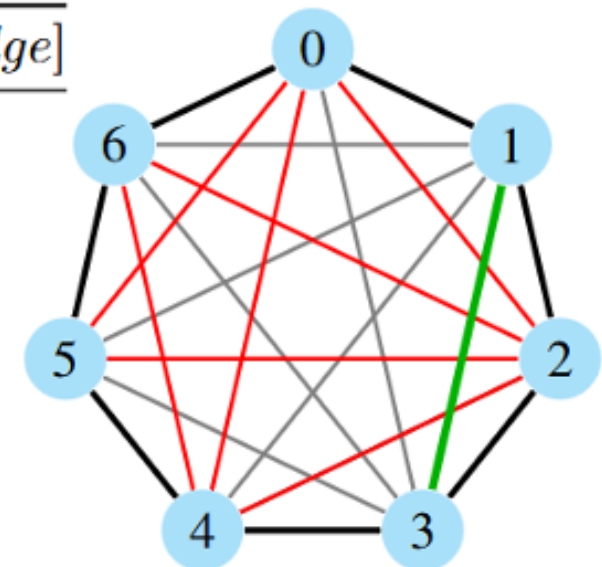
Moyal, P., Bušić, A., and Mairesse, J. (2021). A product form for the general stochastic matching model. Journal of Applied Probability, 58(2):449-468



(a)

$Edge$	$\mathbb{E}[G + Edge]$	$Edge$	$\mathbb{E}[G + Edge]$
$\emptyset$	27.8	$\{1, 6\}$	22.5
$\{1, 3\}$	7.8	$\{3, 6\}$	22.5
$\{0, 2\}$	28.8	$\{2, 4\}$	28.4
$\{0, 3\}$	27.6	$\{2, 5\}$	29.2
$\{0, 4\}$	27.9	$\{2, 6\}$	29.2
$\{0, 5\}$	28.7	$\{3, 5\}$	12.8
$\{1, 4\}$	27.1	$\{4, 6\}$	28.9
$\{1, 5\}$	11.9		

(b)



(c)

- Under FCFM policy, performance can worsen if bad edges are added.
- The same holds for *all* greedy policies.

Bušić, A., Fourneau, J.-M., Lunven, A., and Li, S. (2025). Performance paradoxes in matching systems are not that rare. In International Conference on Network Games, Artificial Intelligence.

Bušić, A., Cadas, A., Doncel, J., and Fourneau, J.-M. (2024). Performance paradox of dynamic matching models under greedy policies. Queueing Systems, 107(3):257–293.

$$C = 1 + \sum_{U\text{-independent}} \sum_{\sigma \in \mathcal{S}(|U|)} \prod_{j=1}^{|U|} \frac{\mu(u_{\sigma(j)})}{\mu(N(\{u_{\sigma(1)}, \dots, u_{\sigma(j)}\})) - \mu(\{u_{\sigma(1)}, \dots, u_{\sigma(j)}\})}$$

$$\pi(w) = \frac{1}{C} \prod_{i=1}^d \frac{\mu(w_i)}{\mu(N(\{w_1, \dots, w_i\}))}$$

> 0?

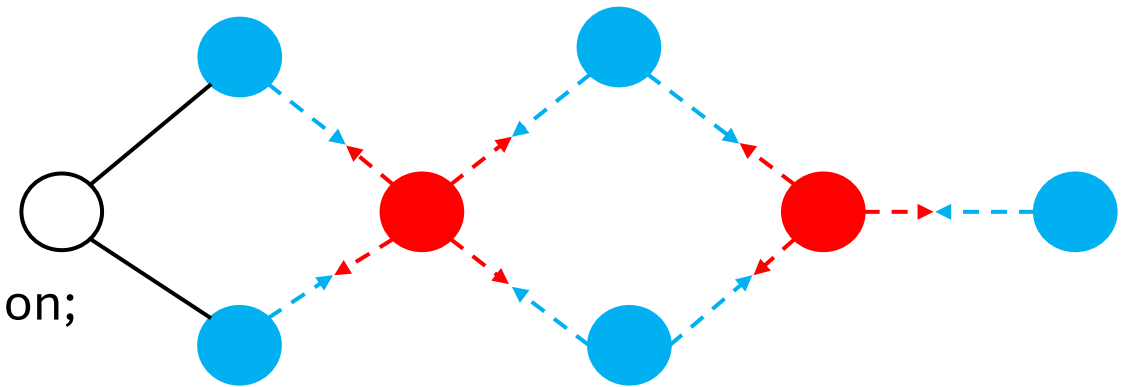
Stability criterion: (G, μ) is stabilized by *some policy*
iff for all independent sets U ,

$$\mu(U) < \mu(N(U))$$

- Necessity: law of large number
- Sufficiency:
 - negative drift for quadratic Lyapunov function;
 - First-Come-First-Matched, Match-the-Longest.

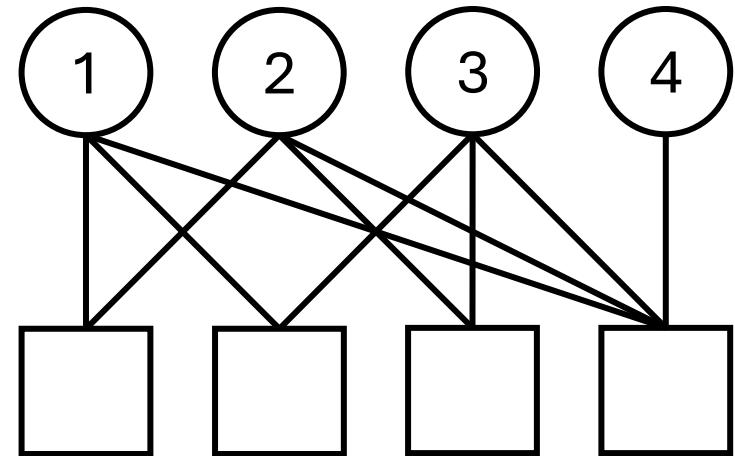
These are greedy policies.

What about *hypergraphs*?



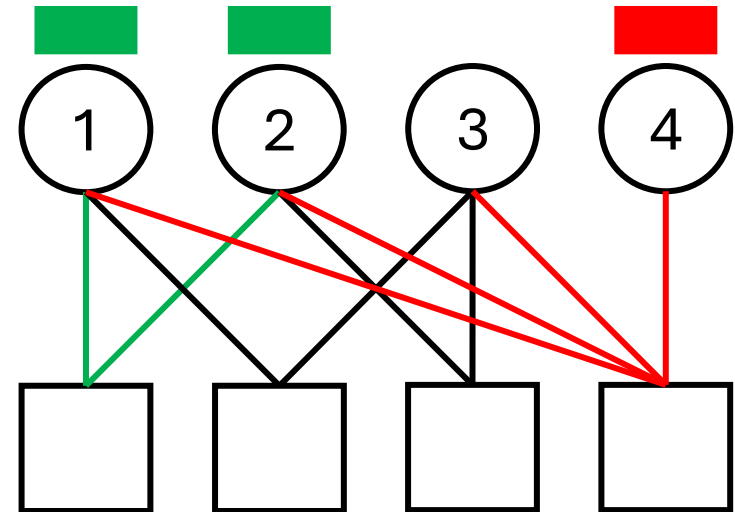
Mairesse, J. and Moyal, P. (2016). Stability of the stochastic matching model. Journal of Applied Probability, 54 (3), pp.1064-1077.

- A **hypergraph** $G = (V, E)$:
 - Vertices: item classes $\bigcirc$;
 - Edges: matching types $\square$.
- Distribution μ on V .
- At time t , a class- v item arrives to the buffer i.i.d. according to μ .
- If **one item for each class u_i** are in the buffer, and $\{u_1, \dots, u_k\} \in E$, they *can* be combined and form a type- e matching, then leave the buffer together.
- Motivation: multi-party kidney donation.
 - Three-party donation decreases waiting time (Roth, Sönmez, and Ünver, 2007).
- Other applications: multi-stop ride hailing, multi-party good exchange.



Roth, A.E., Sönmez, T., and Utku Ünver, M (2007). Efficient kidney exchange: Coincidence of wants in markets with compatibility-based preferences. *American Economic Review*, 97(3):828-851.

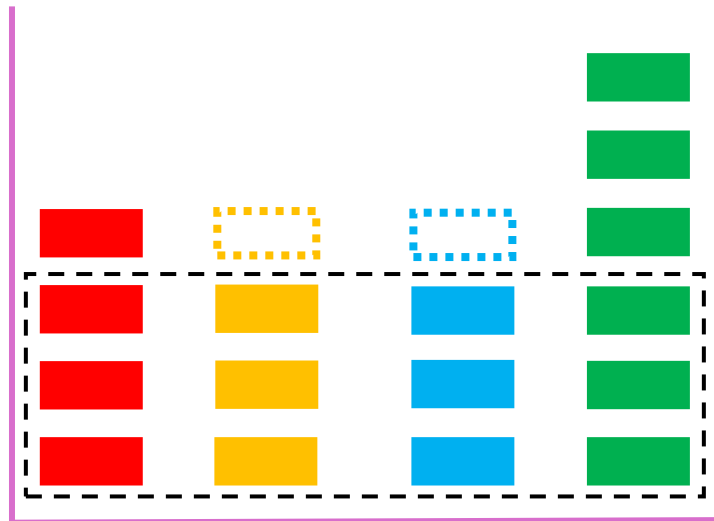
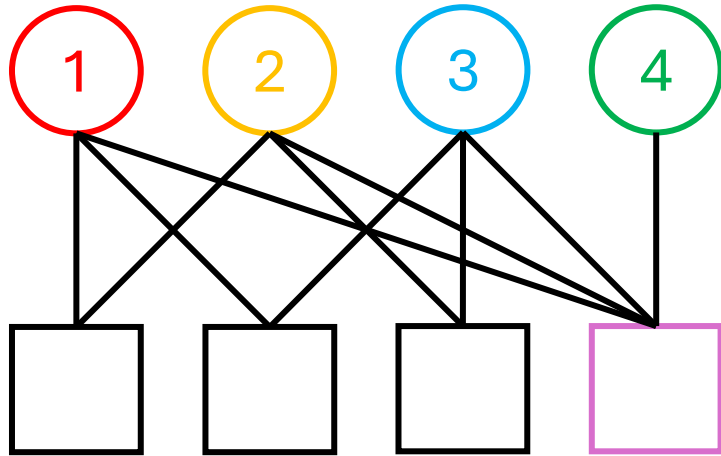
- But greedy policies are *not* maximally stable.
- Problem: given (G, μ) ,
 - What are the necessary and sufficient criteria for stability?
 - If it is stabilisable, find a stabilising policy.
- Question: what is a good alternative to greedy policies?



1. Online assignment rate: $\nu = (\nu_{u,e})_{u \in e, e \in E}$
2. Online assignment policy:
 - At time t , choose a rate ν .
 - If vertex u is drawn, assign the newly arrived item to type e with probability $\frac{\nu_{u,e}}{\mu_u}$.
 - **This item only leaves the buffer via a type- e matching.**
3. A support U is $U = \{(u, e) | q_{u,e} > 0\}$ for some buffer state $q = (q_{u,e})_{u \in e, e \in E}$.

Stability criterion: (G, μ) is stabilizable iff **for all supports U** , there exists an online assignment rate ν^U (depending on U) such that for all $(v, e) \in U$,

$$\nu_{v,e}^U < \frac{1}{|e|} \sum_{u \in e} \nu_{u,e}^U$$



Example:

- Support $U \supseteq \{(1, e), (4, e)\}$.
- To have a matching, we need a class-2 and a class-3 item.
- We need them to arrive more often than class-1 and class-4 items to balance out the buffer.
- The stability criterion now reads:

$$\lambda_{e, \blacksquare} < \frac{\lambda_{e, \blacksquare} + \lambda_{e, \blacksquare} + \lambda_{e, \blacksquare} + \lambda_{e, \blacksquare}}{4}$$

and

$$\lambda_{e, \blacksquare} < \frac{\lambda_{e, \blacksquare} + \lambda_{e, \blacksquare} + \lambda_{e, \blacksquare} + \lambda_{e, \blacksquare}}{4}$$

- How to quantify the imbalance?

$$\mathcal{L} = (\blacksquare_e - \blacksquare_e)^2 + (\blacksquare_e - \blacksquare_e)^2 + (\blacksquare_e - \blacksquare_e)^2 + (\blacksquare_e - \blacksquare_e)^2 + \dots$$

- **Sufficiency:** let $X(t) = q$,

- Lyapunov function: $\mathcal{L}(q) = \sum_{e \in E} \sum_{\{u,v\} \in e^2} (q_{u,e} - q_{v,e})^2$.
- For some constant C_0 , given an assignment rate ν

$$\mathbb{E}[\Delta \mathcal{L} | X(t) = q] \leq C_0 + \sum_{(v,e) \in U} q_{v,e} |e| \left(\nu_{v,e} - \frac{1}{|e|} \sum_{u \in e} \nu_{u,e} \right) \leq -\varepsilon$$

- Stability criterion gives negative drift for all $\|q\|_1$ large enough.

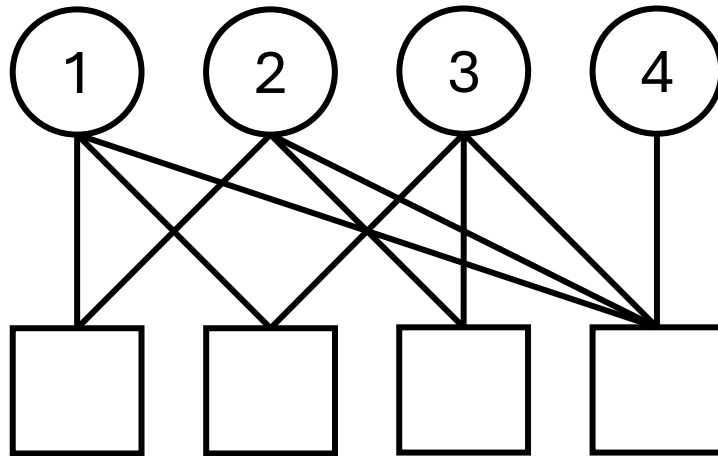
- **Necessity:** suppose the criterion fails for a support U ,

- Define $Y(t) = (Y_{u,e}(t))_{(u,e) \in U}$ by

$$Y_{v,e}(t) = X_{v,e}(t) - \frac{1}{|e|} \sum_{u \in e} X_{u,e}(t)$$

- X is a Markov chain, thus so is Y
- $X(\tau) = 0 \Rightarrow Y(\tau) = 0$.
- (G, μ) stabilisable $\Leftrightarrow \mathbb{E}[\tau] < \infty \Rightarrow Y$ positive recurrent.
- But the criterion fails, Y cannot be positive recurrent.

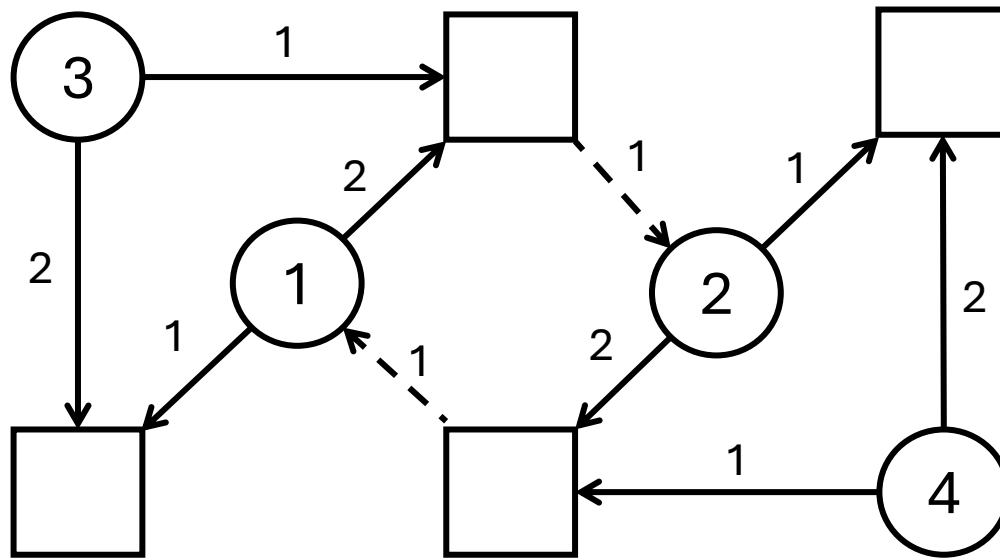
- Incidence matrix $A \in \{0, 1\}^{V \times E}$ of G : $A_{u,e} = 1$ iff $u \in e$.
- Stability criterion using A :
 - (Comte et al., 2021) When G is a graph, (G, μ) is stabilisable iff
 - A is surjective, and
 - There exists $\lambda \in \mathbb{R}_{\geq 0}^E$ such that $A\lambda = \mu$.
 - (N., Bušić, 2026) The same holds when G is a hypergraph.



$$A = \begin{pmatrix} 1 & 1 & 0 & 1 \\ 1 & 0 & 1 & 1 \\ 0 & 1 & 1 & 1 \\ 0 & 0 & 0 & 1 \end{pmatrix} \quad \det A = 2$$

Comte, C., Mathieu, F., Varma, S. M., and Bušić, A. (2025). Online Stochastic Matching: A Polytope Perspective. arXiv:2112.14457
 Nguyen, D.D., and Bušić, A. (2026). Stability in stochastic hypergraph matching: necessary and sufficient criteria. To appear.

- Some generalisations:
 - Non-Markovian policies;
 - Reinterpreting A as matching matrix;
 - Abandonment, service time;
 - Probabilistic matching.
- Applications:
 - Matching with multiplicities;
 - Routing, production line.



$$A = \begin{pmatrix} 1 & 0 & 2 & -1 \\ 0 & 1 & -1 & 2 \\ 2 & 0 & 1 & 0 \\ 0 & 2 & 0 & 1 \end{pmatrix}$$

“Greedy policies are bad.”
- Georgios Paschos